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Take you to know the Tibetan adolescents: Their main social relationships and the effect on the prosocial behavior

Ming Kong *, Yahua Lu

Shanghai Jiao Tong University School of Education, Shanghai 200240, China

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ABSTRACT

Tibetan adolescents in the Tibet Autonomous Region are gradually becoming the mainstay of the development and construction of the Tibet Autonomous Region and the construction of a harmonious society, and it is necessary to explore the positive attitudes and behaviors of this group. For this reason, this study collected 544 valid teacher-student paired questionnaires through the questionnaire survey method, and based on social support theory, explored the influence of the main social relationships (parent–child, teacher–student, and peer relationships) of Tibetan adolescents on their prosocial behaviors as well as the mechanism underlying this influence. The results showed that (1) the main social relationships of Tibetan adolescents were unique; (2) the parent–child, teacher–student, and peer relationships of Tibetan adolescents had significant positive effects on their prosocial behaviors and the teacher-student relationship had the greatest effect; (3) social support mediated the relationship between parent–child, teacher–student, and peer relationships of Tibetan adolescents, on the one hand, and their prosocial behaviors on the other; and (4) the psychological resilience of Tibetan adolescents positively moderated the relationship between their parent–child, teacher–student, and peer relationships and social support, while psychological resilience had a positive moderating effect on the mediating effect. This study conducts empirical research based on detailed data on Tibetan adolescents to provide scientific guidance for stimulating positive behaviors in this group. Moreover, based on the social support theory, it helps to understand the mechanisms of the main social relationships of Tibetan adolescents on their prosocial behaviors.

1. Introduction

Data from the “Major Data Bulletin of the Seventh China Population Census of the Tibet Autonomous Region (TAR)”[1] show that the resident population of the region is 3,648,100, of whom 3,137,901 are Tibetans, accounting for 86.1% of the region’s resident population, this demonstrates that the resident population of the TAR continues to be mainly composed of Tibetans. Furthermore, in the ten years between the sixth (2010) and seventh (2020) China population censuses, the proportion of people with secondary education in the TAR rose from 17.2% to 22.8% of the region’s resident population, an increase of about 6 percentage points. It can be seen that this group of Tibetan adolescents who are receiving or have completed their secondary school studies is gradually becoming the mainstay of the construction of the TAR.

Existing studies on Tibetan adolescents in the TAR have focused mostly on geographical features, such as the negative effects of high altitude on individual psychology (DelMastro et al., 2011; Ishikawa et al., 2016), but significantly less attention has been given to the positive attitudes or behaviors of this group, such as prosocial behavior. Prosocial behavior refers to voluntary actions or tendencies that benefit others and society in social interactions, including sharing, cooperating, helping, and caring (Eisenberg et al., 2006). Numerous studies have shown that prosocial behavior has a positive impact on adolescents’ academic performance, interpersonal relationships, and social adjustment functions (Carlo et al., 2018; Layous et al., 2012; Memmott-Elison et al., 2020). Therefore, considering the important role of prosocial behavior in the development of individuals and social harmony, especially in the process of promoting the unification and social stability of a multiethnic country, it is highly valuable to explore the influencing factors and mechanisms of the prosocial behavior of Tibetan adolescents.

The existing research on the prosocial behavior of adolescents is mainly based on the perspectives of attachment theory, social learning theory, and self-determination theory and explores the influence of

* Corresponding author.
E-mail addresses: kongming@sjtu.edu.cn (M. Kong), luyahua000@sjtu.edu.cn (Y. Lu).

[1] https://tjj.xizang.gov.cn/xxgk/tjxx/tjgb/202105/t20210520_202889.html.

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parent–child, teacher–student, or peer relationships on this prosocial behavior (Froiland et al., 2019; Roorda et al., 2011; van Hoorn et al., 2016; Yoo et al., 2013). These studies have laid a solid theoretical foundation for researching Tibetan adolescents. However, the existing research often focuses only on a single kind of social relationship and has not yet comprehensively examined the impact of multiple kinds of social relationships (i.e., individuals are at the center of the intersection of multiple relationships in social networks) on the prosocial behavior of adolescents from the perspective of relational integration. At the same time, adolescents are at an age when they are undergoing a sensitive period for establishing social relationships with family members and significant others at school (Boele et al., 2019; Obsuth et al., 2017). Social support from parents, teachers, and peers during this period provides a basis for increasing the frequency of prosocial behaviors among adolescents (Hu et al., 2021; Kristofferson et al., 2014), who can obtain social support from different types of social relationships (lzaguirre et al., 2021; Watson et al., 2019). Therefore, this study proposes to introduce parent–child, teacher–student, and peer relationships into the model simultaneously to explore the impact of these three main social relationships on the prosocial behavior of Tibetan adolescents and to compare the magnitude of the effects of each type of relationship to provide a more focused basis and suggestions for educational management and student growth.

According to social support theory, the support that individuals derive from others mainly includes emotional, informational, instrumental, and belonging support (Cohen & Hoberman, 1983; Cutrona & Russell, 1990). Specifically, emotional support refers to the care, understanding, respect, or love that an individual receives from others; information support involves providing individuals with various types of information that can be used to solve problems, such as advice, guidance, or recommendations; instrumental support is primarily about providing help such as resources, services, or assistance to individuals; and belonging support comes from establishing relationships with others or playing a role in an organization to satisfy an individual's sense of social belonging (Cohen & Wills, 1985). Based on this theory, social support, as an external protective resource, is closely related to the quality and function of interpersonal relationships, and high-quality relationships can produce positive changes in an individual's life and ultimately promote positive behavioral responses (Arslan, 2017; Hobfoll et al., 1990). Therefore, we propose to use social support as a mediating variable to explain the mechanisms connecting primary social relationships and the prosocial behaviors of Tibetan adolescents. At the same time, this study also analyzes the role of psychological resilience in this process and proposes to examine its moderating effect on the main social relationships to Tibetan adolescents' social support, as well as its moderating effect on the mediating role.

1.1. Parent–child, Teacher–student, and Peer Relationships and Prosocial Behavior

Social support theory suggests that the size and cohesiveness of an individual's social relationship network and the type of relationships in that network are critical to seeking or receiving social support (Thoits, 1995). The closer the relationships between individuals and significant others in their social environment are, the more able they are to cope with external pressures, maintain positive emotions, and generate positive behaviors (Barrera, 1986). For adolescents, parent–child, teacher–student, and peer relationships are the three most important social relationships that have a significant impact on the formation and development of their prosocial behavior (Eisenberg et al., 2015; Jiang et al., 2018).

Specifically, in a close parent–child relationship, the parent's prosocial precepts and deeds are more easily accepted and imitated by the child (Michiels et al., 2008), and a high-quality parent–child relationship helps students pay more attention to the needs and feelings of others, understand more about caring for and helping others (Boele et al., 2019; Clark & Ladd, 2000), and effectively improve their probability of students' positive behaviors (Eisenberg et al., 2010; Waller et al., 2014). Moreover, the quality of a teacher–student relationship is an important predictor of student behavior adjustment (Obsuth et al., 2017). A harmonious teacher–student relationship plays an important role in dealing with conflicts within the class, guiding students to behave correctly and preventing them from engaging in incorrect behavior (Longobardi et al., 2020), which can effectively reduce the probability of their negative behaviors (Jungert et al., 2016; Marengo et al., 2018). Furthermore, teachers who maintain high-quality relationships with their students often foster a positive sense of community by encouraging students to share knowledge and skills, thereby promoting cooperation among students (Mikami et al., 2011), and these interactions can effectively encourage students to internalize teachers' encouragement of prosocial behaviors and incorporate them into their practical actions (Chapman et al., 2013). Additionally, as autonomy and independence increase during adolescence, the impact of peer relationships on adolescents is crucial (Rubin et al., 1998), especially when they are faced with stressful events. High-quality peer relationships can effectively alleviate the negative effects of stress on emotions and behavior and have an important protective effect on student growth (Mounts et al., 2006; Van & Van, 2015). At the same time, high-quality peer relationships help students share knowledge, skills, and resources among their peers and improve their awareness and level of serving others (Stern & Cassidy, 2018; van Hoorn et al., 2016). With all this considered, we hypothesize the following:

H1: The parent–child (H1a), teacher–student (H1b) and peer relationships (H1c) of Tibetan adolescents are significantly and positively correlated with their prosocial behavior.

1.2. The mediating role of social support

Social support is the general or specific support and help that individuals receive from others or from social networks (Cohen & Wills, 1985), mainly including emotional, informational, instrumental, and belonging support (Cohen & Hoberman, 1983; Cutrona & Russell, 1990). Social support theory suggests that interpersonal relationships are the basis for giving and receiving social support and that high-quality interpersonal relationships can provide individuals with a wider range of support types (Gottlied & Bergen, 2010). For adolescents, high-quality parent–child, teacher–student, and peer relationships can enhance their perception of social support, thus increasing the frequency of their prosocial behavior.

Specifically, a high–quality parent–child relationship can enable students to feel loved, cared for, and understood (Malecki & Demaray, 2003). In this high-quality interaction process, parents can keenly perceive children's emotional and psychological needs and provide timely appropriate support and help so that children can feel emotional and belonging support (Mesurado & Richaud, 2017; Michiels et al., 2008). This sense of support facilitates the development of children's empathy abilities and lays the foundation for a more positive connection with others (Miklikowska et al., 2011; Zhou et al., 2022). Furthermore, teachers play an important role in the growth of students. In a quality teacher–student relationship, teachers express respect and appreciation, provide care and attention, and offer support and assistance to students. Students who perceive teachers' support and care are often more willing to adopt various types of support provided by teachers and also more inclined to show the behaviors that they expect and advocate (Obsuth et al., 2017; Wang & Eccles, 2012), such as helpfulness, understanding, and solidarity. At the same time, in a quality teacher–student relationship, the emotional, informational, instrumental, and belonging support given to students by teachers can also effectively improve students' positive attitude and participation in organizational construction and personal growth (Longobardi et al., 2020; Ruzek et al., 2016). In addition, in the adolescent stage, high-quality peer relationships are more likely to enable students to feel a sense of belonging to the group, making them feel recognized, accepted, included, and trusted (Boele et al., 2019; Wang & Eccles, 2012), which can meet their psychological needs (Gorrese, 2016). This kind of support, encouragement, and companionship from peers promotes mutual help among peers (Farrell et al., 2017; van Hoorn et al., 2016). With all this considered, we hypothesize the following:

H2: The parent–child (H2a), teacher–student (H2b) and peer relationships (H2c) of Tibetan adolescents have an affect on their prosocial behavior via social support.

1.3. The moderating role of psychological resilience

Psychological resilience is a relatively stable personality trait (Block & Kremen, 1996) and an important predictor of an individual's mental health and development status (He et al., 2014). Individual differences based on traits and experiences are an important class of self-regulatory resources that often mitigate the negative consequences of adverse events (De Clercq & Belausteguigoitia, 2017). Research has shown that the way and extent to which individuals interact with others may depend on their personal characteristics such as psychological resilience (Meng et al., 2019), which can help improve interpersonal adjustment and mitigate relationship conflict (Block & Kremen, 1996). Specifically, adolescents with high psychological resilience usually have more positive emotional experiences when faced with problems in their parent–child, teacher–student, or peer relationships, which enables them to actively self-adjust and flexibly respond in adverse situations and find solutions to difficulties or challenges (Hu et al., 2022). Studies have confirmed that individuals are more likely to establish strong social support networks or perceive social support when they have sound personality traits (Barrera, 1986; Thoits, 1995). Thus, the influence of parent–child, teacher–student, and peer relationships on the social support of Tibetan adolescents is likely to be moderated by their psychological resilience.

Furthermore, students with high levels of psychological resilience are more self-accepting and are able not only to hold positive attitudes toward various aspects of their past, present, and future (Huang et al., 2020; Xu et al., 2017) but also to combine internal personal resources with social relationships and to seek necessary social support from parents, teachers, or peers (Sisto et al., 2019; Xue et al., 2022). Therefore, the higher the psychological resilience of Tibetan adolescents is, the more likely they are to effectively build and maintain their surrounding social relations, thereby obtaining more social support and maintaining a more positive psychological state. On the other hand, students with low levels of psychological resilience have lower self-acceptance and are more likely to develop feelings of inferiority (Chen et al., 2017; Huang et al., 2020), which reduces not only their interactions with parents, teachers, and peers but also the willingness of others to provide them with social support (Ai & Hu, 2016; Fletcher & Sarkar, 2013). In addition, students with high levels of psychological resilience are more likely to have an unyielding spirit and an attitude of continuous struggle (Bonanno, 2004; Sisto et al., 2019), which are highly regarded in Chinese traditional cultural values (Lan & Wang, 2019), prompting parents, teachers, and peers to give them more positive evaluations and provide more emotional, informational, instrumental, and belonging support to the best of their ability (Wu et al., 2014; Pinkerton & Dolan, 2007). With all this considered, we hypothesize the following:

H3: The psychological resilience of Tibetan adolescents positively moderates the relationship between their parent–child (H3a), teacher–student (H3b), and peer relationships (H3c) and social support.

1.4. Moderated mediating effect

Combining Hypotheses 2 and 3, we further infer that the mediating effect of social support between parent–child, teacher–student, and peer relationships, on the one hand, and prosocial behavior, on the other hand, of Tibetan adolescents may also be moderated by psychological resilience. Specifically, when Tibetan adolescents have higher levels of psychological resilience, they are more likely to maintain positive attitudes and behaviors, and others' willingness to provide them social support increases (Fletcher & Sarkar, 2013; Xue et al., 2022). When adolescents receive more social support, they show more positive prosocial behavior in return for help from others (Kristofferson et al., 2014; Staub & Vollhardt, 2008). Conversely, when Tibetan adolescents have lower levels of psychological resilience, they have diminished perceptions of self-worth (Lan & Wang, 2019; Mesurado & Richaud, 2017) and show more negative attitudes and behaviors, which is not conducive to maintaining close relationships with parents, teachers, and peers, and their perception of social support also decreases (Gottlied & Bergen, 2010; Salovey et al., 2000), which reduces their prosocial behaviors.

H4: The indirect effects of parent–child, teacher–student, and peer relationships on the prosocial behavior of Tibetan adolescents via social support are positively moderated by psychological resilience.

1.5. Current study

Based on the above analysis, this research aims to empirically explore the influence of the main social relationships (parent–child, teacher–student, and peer relationships) of Tibetan adolescents on their prosocial behaviors as well as the mechanism underlying this influence from the perspective of social support theory. The purpose of this examination is as follows: first to clarify whether the primary social relationships of Tibetan adolescents can positively influence their prosocial behavior; second, to reveal the mechanism by which the parent–child, teacher–student, and peer relationships of Tibetan adolescents affect their prosocial behavior through the pathway of social support; and third, to explore the boundary conditions under which Tibetan adolescents' primary social relationships act on their prosocial behavior from the perspective of psychological resilience. To achieve the above research purposes in this study and to supplement evidence to stimulate prosocial behavior among Tibetan adolescents, an empirical analysis based on detailed data from these adolescents in the TAR is conducted. In this study, the research model shown in Fig. 1 is constructed.

2. Methods

2.1. Overview of the study

Before answering the question of whether and how the main social relationships (parent–child, teacher–student, and peer relationships) among Tibetan adolescents affect their prosocial behavior, we would like to first clarify whether there are significant differences between Tibetan adolescents in the TAR, Han adolescents in the TAR, and Han adolescents in mainland China in terms of parent–child, teacher–student, and peer relationships. To this end, the research design consisted of two studies: (1) Study 1 aims to examine the uniqueness of the main social relationships among Tibetan adolescents in the TAR; (2) Study 2 intends to explore the influence of the main social relationships of Tibetan adolescents on their prosocial behaviors as well as the mechanism underlying this influence from the perspective of social support theory.

2.2. Study 1

2.2.1. Sample and procedures

In order to examine the uniqueness of the main social relationships among Tibetan adolescents, three groups of samples, this study selected three groups of samples from Tibetan adolescents in the TAR, Han adolescents in the TAR and Han adolescents in mainland China for comparison, as shown in Table 1:

Sample of Tibetan adolescents in the TAR: The research was

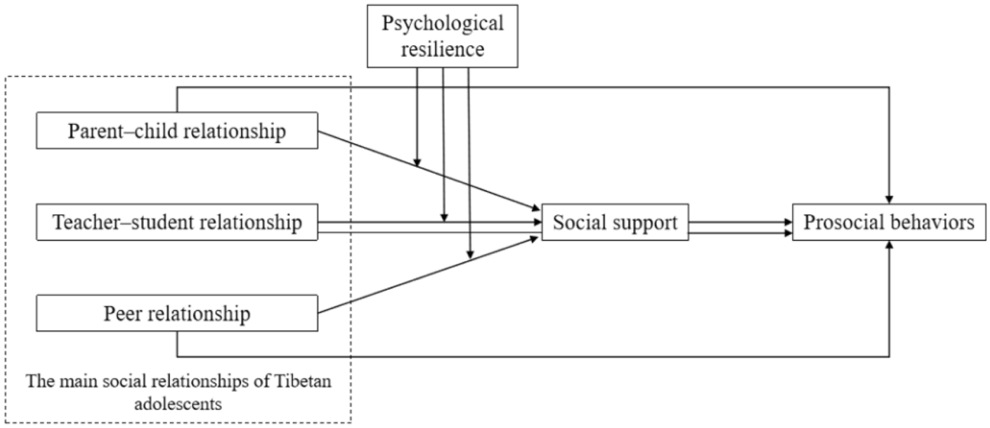

Fig. 1. Research model.

Table 1
Study 1 sample information.

Sample	Sample size	Boys	Girls	Average age
Tibetan adolescents in the TAR	254	100	154	16.1 (*SD* = 0.80)
Han adolescents in the TAR	208	117	91	16.2 (*SD* = 1.05)
Han adolescents in mainland China	470	268	202	15.7 (*SD* = 1.84)

conducted in Naqu Middle School and Lhasa Middle School, a total of 254 participants were selected by cluster sampling with class as unit, with a mean age of 16.1 years ($SD = 0.80$), of which 100 were boys and 154 were girls.

Sample of Han adolescents in the TAR: The research was conducted in the Tibetan Military Region Lhasa 8–1 School, Linzhi Middle School and Lhasa Middle School, a total of 208 participants were selected by cluster sampling with class as unit, with a mean age of 16.2 years ($SD = 1.05$), of which 117 were boys and 91 were girls.

Sample of Han adolescents in mainland China: Research was conducted at two high schools in Wuhan and Chengdu, a total of 470 participants were selected by cluster sampling with class as unit, with a mean age of 15.7 years ($SD = 1.84$), of which 268 were boys and 202 were girls.

Data for Study 1 were collected in school. After obtaining the informed consent of the teachers, participants, and their parents, when the investigation was administered, the class was taken as the unit, and the participants filled out a self-report questionnaire during the agreed self-study class time and collected it on the spot. Before participants filled out the questionnaire, they were informed about voluntary participation and the confidentiality and anonymity of the data.

2.2.2. *Study 1 measures*

Parent–child relationship: The scale was revised by Driscoll and Pianta (2011) based on the Parent–Child Relationship Scale developed by Pianta (1992) and contains 15 questions (e.g., "I share an affectionate, warm relationship with my parents" and "I get angry with my parents easily"). Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by the students. In a comparative study between Tibetan adolescents in the TAR and Han adolescents in the TAR, the Cronbach'α value of this scale was 0.70; in a comparative study of Tibetan adolescents in the TAR and Han adolescents in mainland China, the Cronbach'α value of this scale was 0.72.

Teacher–student relationship: Pianta's (2001) Teacher–Student Relationship Scale was used, which contains 15 items and is currently the most accepted scale for measuring teacher-student relationships (e.g., "If upset, I will seek comfort from my teacher" and "I remain angry or is resistant after being disciplined by my teacher"). Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by the students. In a comparative study between Tibetan adolescents in the TAR and Han adolescents in the TAR, the Cronbach'α value of this scale was 0.70; in a comparative study of Tibetan adolescents in the TAR and Han adolescents in mainland China, the Cronbach'α value of this scale was 0.79.

Peer relationship: The eighteen items developed by Marsh et al. (1984) were used to measure peer relationship (e.g., "I can easily become friends with boys" and "I can easily become friends with girls"). Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by the students. In a comparative study between Tibetan adolescents in the TAR and Han adolescents in the TAR, the Cronbach'α value of this scale was 0.82; in a comparative study of Tibetan adolescents in the TAR and Han adolescents in mainland China, the Cronbach'α value of this scale was 0.83.

2.2.3. *Study 1 results*

To examine the uniqueness of the main social relationships of Tibetan adolescents in the TAR, this study used the independent samples *t*-test to analyze three groups of samples: Tibetan adolescents in the TAR, Han adolescents in the TAR, and Han adolescents in mainland China.

First, we examined whether there were significant differences in parent–child, teacher–student and peer relationships between Tibetan adolescents in the TAR and Han adolescents in the TAR. As shown in Table 2: (1) In terms of parent–child relationships, there were significant differences between Tibetan adolescents in the TAR ($M = 1.62$, $SD = 1.32$) and Han adolescents in the TAR ($M = 0.79$, $SD = 0.96$), $t(454) = 7.88$, $p < 0.001$. (2) In terms of teacher–student relationships, there were no significant differences between Tibetan adolescents in the TAR ($M = 1.14$, $SD = 1.04$) and Han adolescents in the TAR ($M = 1.11$, $SD = 0.87$), $t(460) = 0.40$, $p > 0.05$. (3) In terms of peer relationships, there were

Table 2
Comparison of differences in main social relations between Tibetan adolescents in the TAR and Han adolescents in the TAR ($M \pm SD$).

Variables	Tibetan in the TAR	Han in the TAR	*t*	*P*
parent–child relationships	1.62 ± 1.32	0.79 ± 0.96	7.88***	0.000
teacher–student relationships	1.14 ± 1.04	1.11 ± 0.87	0.40	0.690
peer relationships	4.01 ± 0.51	4.45 ± 0.79	−6.95***	0.000

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

significant differences between Tibetan adolescents in the TAR (M = 4.01, SD = 0.51) and Han adolescents in the TAR (M = 4.45, SD = 0.79), $t(341) = -6.95$, $p < 0.001$.

Second, we examined whether there were significant differences in parent–child, teacher–student and peer relationships between Tibetan adolescents in the TAR and Han adolescents in mainland China. As shown in Table 3: (1) In terms of parent–child relationships, there were significant differences between Tibetan adolescents in the TAR (M = 1.62, SD = 1.32) and Han adolescents in mainland China (M = 0.72, SD = 0.77), $t(405) = 9.44$, $p < 0.001$. (2) In terms of teacher–student relationships, there were significant differences between Tibetan adolescents in the TAR (M = 1.14, SD = 1.04) and Han adolescents in mainland China (M = 0.84, SD = 0.61), $t(408) = 4.11$, $p < 0.001$. (3) In terms of peer relationships, there were significant differences between Tibetan adolescents in the TAR (M = 4.01, SD = 0.51) and Han adolescents in mainland China (M = 4.34, SD = 0.76), $t(340) = -5.33$, $p < 0.001$.

2.2.4. Study 1 discussion

Study 1 verified that there are significant differences in the parent–child, teacher–student, and peer relationships of Tibetan adolescents in the TAR, Han adolescents in the TAR and Han adolescents in mainland China. Specifically, parent–child relationships are significantly stronger among Tibetan adolescents in the TAR than among Han adolescents in the TAR and Han adolescents in mainland China, and teacher-student relationships are significantly stronger among Tibetan adolescents in the TAR than among Han adolescents in mainland China. However, peer relationships are significantly weaker among Tibetan adolescents in the TAR than among Han adolescents in the TAR and Han adolescents in mainland China. The main reasons for these differences may be as follows. First, The local traditional culture has a profound influence on the Tibetan cultural system. Although the modern educational system has become more widespread, the notion of a traditional Tibetan upbringing continues to have a profound influence on the growth of the younger generation. In accordance with local traditional culture, children are instructed from an early age to exhibit filial piety toward their parents and to pursue family harmony, which becomes the basis of the belief of Tibetan families in close relationships between parents and children (Chen, 2015). Moreover, most Tibetan adolescents have lived with their parents day and night since they were young, and the main method of education in this context is family education (Shi et al., 2019). Therefore, Tibetan adolescents have relatively close relationships with their parents. Second, according to the boarding school model of middle school in TAR, teachers may be important objects of attachment for adolescents in the TAR aside from their parents. Furthermore, Tibetan traditional culture strongly emphasizes respect for one's teachers. Students always use honorifics when addressing teachers, and parents have great trust in teachers' discipline (Chen, 2015). Therefore, teacher–student relationships among Tibetan adolescents are relatively close. Third, Tibetan adolescents are more concerned with what others think about them and are very conscious of the impact of their words and actions on the emotions of others, to the point that they even refrain from expressing their grievances or disagreements with others (Chen, 2017). Simultaneously, close interpersonal interaction may result in psychological stress for their peers. Considering the psychological condition of their peers, Tibetan adolescents have a strong sense of boundaries in the context of interacting with their peers (Chen, 2015), which may cause their peer relationships to be significantly weaker than peer relationships among Han students in the TAR and Han students in mainland China. The above differences lay the foundation for the discussion on the main social relations of Tibetan adolescents in the TAR.

Table 3
Comparison of differences in main social relations between Tibetan adolescents in the TAR and Han adolescents in mainland China ($M \pm SD$).

Variables	Tibetan in the TAR	Han in mainland China	t	P
parent–child relationships	1.62 ± 1.32	0.72 ± 0.77	9.44***	0.000
teacher–student relationships	1.14 ± 1.04	0.84 ± 0.61	4.11***	0.000
peer relationships	4.01 ± 0.51	4.34 ± 0.76	−5.33***	0.000

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

2.3. Study 2

2.3.1. Sample and procedures

This study took Tibetan adolescents in the TAR as the research object and conducted a questionnaire survey in Ali Middle School, Naqu Middle School, and Lhasa Middle School in the TAR. To increase the rigor of the research design, this study adopted multiple time points, a multi-data source research approach. To avoid the problem of common method bias, data collection was conducted by a paired teacher-student survey method. Specifically, data were collected at two time points: at time point 1, parent–child relationships, teacher-student relationships, peer relationships, and demographic information were self-assessed by the researched students, and 1,200 questionnaires were distributed, with 1,132 returned (recovery rate of 94.3%). At time point 2, one month later, the student and his or her classroom teacher were researched, the student was asked to self-assess perceived social support and psychological resilience, and the student's classroom teacher was asked to evaluate the student's prosocial behavior. Student questionnaires were distributed to 1132, with 961 returned (recovery rate of 84.9%); teacher questionnaires were distributed to 1132, with 617 returned (recovery rate of 54.5%). After removing invalid questionnaires with missing values, missing teacher-student pairs, and obvious response tendencies, a total of 544 valid teacher-student paired questionnaires were obtained in this study (effective rate of 88.2%), so the final sample of this study consisted of 544 data from student self-assessments and classroom teacher evaluations. The mean age of the students was 16.5 years (SD = 1.16), of which 278 were boys and 266 were girls.

Data for Study 2 were collected in school, and informed consent was obtained from teachers, participants, and parents for both investigations. When the investigation was administered, cluster sampling was carried out with the class as the unit, and the participants completed the questionnaire during the agreed self-study class time and collected it on the spot. Before participants filled out the questionnaire, they were informed about voluntary participation and the confidentiality and anonymity of the data.

2.3.2. Study 2 measures

Parent–child relationship: The scale was the same as study 1. Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by students at time point 1, with the Cronbach'α value of 0.70.

Teacher–student relationship: The scale was the same as study 1. Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by students at time point 1, with the Cronbach'α value of 0.71.

Peer relationship: The scale was the same as study 1. Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by students at time point 1, with the Cronbach'α value of 0.70.

Social support: The twelve items developed by Zimet et al. (1988) were used to measure social support (e.g., "My friends and family are around when I am in trouble" and "My parents, teachers and classmates care about my feelings"). Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by students at time point 2, with the Cronbach'α value of 0.88.

Psychological resilience: The twenty-five items developed by Wagnild and Young (1993) were used to measure psychological resilience (e.g., “When I make a plan, I will carry it through” and “If someone does not like me, it’s no big deal”). Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was self-assessed by students at time point 2, with the Cronbach’α value of 0.90.

Prosocial behaviors: The five items developed by Goodman et al. (1998) were used to measure prosocial behaviors (e.g., “I try to be nice to other people. I care about their feelings” and “I usually share with others (food, games, pens, etc.”). Answers were given on a 6-point scale ranging from 1 = strongly disagree to 6 = strongly agree. The scale was evaluated by the teacher at time point 2 for students, with the Cronbach’α value of 0.70.

Control variables: According to previous studies, there are differences in the establishment and perception of social relationships among adolescents of different genders (Thoits, 1995; Rose & Rudolph, 2006). Meanwhile, previous studies on prosocial behavior of adolescents have controlled for individual gender differences (Mesurado et al., 2019; Longobardi et al., 2020). Therefore, this study controlled for gender as a control variable, with girls recorded as 0 and boys as 1. In addition, age was controlled for in this study.

2.3.3. *Study 2 results*

SPSS 25.0 and Mplus 8.4 were used for the statistical analysis of the data in this study. First, validation factor analysis was performed using Mplus 8.4 to test the structural and discriminant validity of the variables. Second, preliminary descriptive statistics and correlation analysis tests. Third, Structural Equation Models (SEM) were constructed to test the research hypotheses, and the mediating role in the model and the moderated mediating role hypotheses were tested using a bootstrap approach with repeated sampling as suggested by Preacher and Hayes (2008).

2.3.3.1. Confirmatory factor analysis. As there are many measurement items for individual variables in this study, which may affect the overall fit of the model, scholars suggest packaging the items (Little et al., 2002). Referring to the isolated uniqueness strategy mentioned by Hall et al. (1999), this study used the error correlation method for variables with a unidimensional structure and packaged the items with high correction indices together according to the correction indices of error correlation in the confirmatory factor analysis (CFA) results. For variables with multidimensional structures, each dimension was packaged into a latent factor. After data packaging was completed, this study examined the discriminant validity of six variables: parent–child relationship, teacher–student relationship, peer relationship, psychological resilience, social support, and prosocial behaviors through CFA. The results of the CFA in Table 4 show that the six-factor model fits better compared to the combination of factors in the other models (χ^2/df = 3.66, CFI = 0.98, TLI = 0.96, RMSEA = 0.07, SRMR = 0.06), indicating that the six main variables measured in this study have good discriminant validity.

2.3.3.2. Common method bias. In addition to prosocial behavior, other key variables in this study were measured by self-report of the participants, which may lead to common method bias. To reduce this possible bias, in terms of procedural control, this study used multi-time points (two-stage survey), anonymous measurement, and partial question item reversal. Statistically, the following two methods were used in this study to test for possible common method bias. First, we adopted Harman’s single-factor test (Podsakoff et al., 2003), and the results showed that there were 22 factors with eigenvalues greater than 1, and the cumulative variance explanation rate of the first precipitated factor was 14.80%, not more than 40%. Second, we adopted an unmeasured latent common method factor for testing (Podsakoff et al., 2003). After CFA using Mplus 8.4, a common method factor was added on top of the six-factor model. The results showed that the seven-factor structural model with the addition of the common method factor could not be fitted in Mplus 8.4 software. Together, the results of both methods indicated that the common method bias of the measures was minimal.

Table 4
Results of validation factor analysis.

Models	χ^2	df	χ^2/df	CFI	TLI	RMSEA	SRMR
Six-factor model (PCR; TSR; PR; R; SS; PB)	709.96	194	3.66	0.98	0.96	0.07	0.06
Five-factor model (PCR; TSR; PR; R + SS; PB)	1288.27	199	6.47	0.75	0.71	0.10	0.08
Four-factor model (PCR + TSR + PR; R; SS; PB)	864.85	203	4.26	0.85	0.83	0.08	0.06
Three-factor model (PCR + TSR + PR; R + SS; PB)	1435.95	206	6.97	0.71	0.68	0.11	0.09
Two-factor model (PCR + TSR + PR; R + SS + PB)	1721.64	208	8.28	0.65	0.61	0.12	0.10
One-factor model (PCR + TSR + PR + R + SS + PB)	1740.47	209	8.33	0.64	0.61	0.12	0.10

Note: PCR stands for parent–child relationship, TSR stands for teacher–student relationship, PR stands for peer relationship, R stands for psychological resilience, SS stands for social support, PB stands for prosocial behavior; “+” stands for two factors combined into one factor.

2.3.3.3. Descriptive statistics. The correlations and descriptive statistics of the study variables are presented in Table 5. As expected, parent–child relationships were positively correlated with prosocial behavior (r = 0.12, p < 0.01). Teacher–student relationships were positively correlated with prosocial behavior (r = 0.20, p < 0.01). Peer relationships were positively correlated with prosocial behavior (r = 0.13, p < 0.01). Parent–child relationships were positively correlated with social support (r = 0.48, p < 0.01). Teacher–student relationships were positively correlated with social support (r = 0.41, p < 0.01). Peer relationships were positively correlated with social support (r = 0.51, p < 0.01). Social support relationships were positively correlated with prosocial behavior (r = 0.22, p < 0.01). These results provide a database for further research.

2.3.3.4. Main effects. First, Hypothesis 1a proposed that parent–child relationships of Tibetan adolescents are significantly positively related to their prosocial behavior, as shown in Model 1 of Table 6, after controlling for gender and age, there was a significant positive effect of parent–child relationships on prosocial behavior (b = 0.11, p < 0.05), and Hypothesis 1a was tested. Second, Hypothesis 1b proposed that teacher–student relationships of Tibetan adolescents are significantly positively related to their prosocial behavior, as shown in Model 2 of Table 6, there was a significant positive effect of teacher–student relationships on prosocial behavior (b = 0.18, p < 0.001), and Hypothesis 1b was tested. Finally, Hypothesis 1c proposed that peer relationships of Tibetan adolescents are significantly positively related to their prosocial behavior, as shown in the Model 3 of Table 6, peer relationships had a significant positive effect on prosocial behavior (b = 0.13, p < 0.01), and Hypothesis 1c was tested. Together, Hypothesis 1 was verified. Meanwhile, according to the data results, the teacher–student relationship has the greatest impact on the prosocial behavior of Tibetan adolescents (b = 0.18, p < 0.001), followed by peer relationships (b = 0.13, p < 0.01), with the parent–child relationship being the weakest (b = 0.11, p < 0.05).

Table 5
Means, standard deviations and correlations of study variables.

Variable	*M*	*SD*	1	2	3	4	5	6	7
1. Gender	0.51	0.50							
2. Age	16.53	1.16	0.06						
3. Parent–child relationship	1.82	1.26	−0.07	0.12**					
4. Teacher–student relationship	1.40	1.04	−0.14**	0.10*	0.57**				
5. Peer relationship	4.05	0.54	0.03	−0.00	0.43**	0.43**			
6. Psychological resilience	4.49	0.66	−0.02	0.03	0.08	0.21**	0.14**		
7. Social support	4.55	0.83	−0.09*	0.08	0.48**	0.41**	0.51**	0.15**	
8. Prosocial behavior	5.29	0.58	−0.11**	0.04	0.12**	0.20**	0.13**	0.36**	0.22**

Note: $N = 544$.
** $p < 0.01$.
* $p < 0.05$.

Table 6
Results of main effect and moderating effect analysis.

Variable	Prosocial behavior			Social support		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Gender	−0.11*	−0.09*	−0.12**	−0.06	−0.04	−0.12**
Age	0.04	0.03	0.05	0.03	0.04	0.09*
Parent–child relationship	0.11*			0.46***		
Teacher–student relationship		0.18***			0.39***	
Peer relationship			0.13**			0.49***
Psychological resilience				0.11**	0.08	0.08
Parent–child relationship × Psychological resilience				0.13*		
Teacher–student relationship × Psychological resilience					0.13*	
Peer relationship × Psychological resilience						0.11**
R^2	0.03	0.05	0.03	0.26	0.19	0.30
F	4.84**	9.01***	6.13***	38.78***	25.51***	44.18***

Note: $N = 544$.The data in the table are standardized regression coefficients.
*** $p < 0.001$.
** $p < 0.01$.
* $p < 0.05$.

2.3.3.5. The mediating effect of social support. In this study, Bootstrapping method was used to test for mediating effects by repeated sampling 5000 times through Mplus 8.4 software and the results are shown in Table 7. First, the parent–child relationship of Tibetan adolescents was significantly positively associated with their perceived social support ($b = 0.48$, $p < 0.001$), and social support was also significantly positively associated with their prosocial behaviors ($b = 0.21$, $p < 0.001$). Moreover, the indirect effect of the parent–child relationship on the prosocial behavior of Tibetan adolescents via social support was significant ($b = 0.10$, $p < 0.001$, 95%CI = [0.053, 0.146]). In addition, after adding social support, parent–child relationship had no significant positive effect on prosocial behavior ($b = 0.01$, $p = 0.871$), indicating that Tibetan adolescents' perceived social support mediated the relationship between their parent–child relationship and prosocial behavior, and Hypothesis 2a was verified. Second, the teacher–student relationship of Tibetan adolescents was significantly positively associated with their perceived social support ($b = 0.40$, $p < 0.001$), and social support was also significantly positively associated with their prosocial behaviors ($b = 0.16$, $p < 0.01$). Moreover, the indirect effect of the teacher–student relationship on the prosocial behavior of Tibetan adolescents via social support was significant ($b = 0.07$, $p < 0.01$, 95%CI = [0.028, 0.105]). In addition, after adding social support, teacher–student relationships remained significantly positively associated with prosocial behavior ($b = 0.12$, $p < 0.05$), indicating that Tibetan

Table 7
Mediating effect analysis results.

Variable	Social support			Prosocial behavior		
	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Gender	−0.06	−0.04	−0.11**	−0.10*	−0.08	−0.10*
Age	0.03	0.05	0.09*	0.03	0.02	0.03
Parent–child relationship	0.48***			0.01		
Teacher–student relationship		0.40***			0.12*	
Peer relationship			0.51***			0.04
Social support				0.21***	0.16**	0.19***
R^2	0.24***	0.17***	0.28***	0.06**	0.07**	0.06**
Indirect effects	Model			Effect value	*SE*	95%CI
	Parent–child relationship → Social support → Prosocial behavior			0.10***	0.02	[0.053, 0.146]
	Teacher–student relationship → Social support → Prosocial behavior			0.07**	0.02	[0.028, 0.105]
	Peer relationship → Social support → Prosocial behavior			0.10***	0.03	[0.046, 0.156]

Note: $N = 544$. The data in the table are standardized regression coefficients.
*** $p < 0.001$.
** $p < 0.01$.
* $p < 0.05$.

adolescents' perceived social support mediated the relationship between their teacher–student relationship and prosocial behavior, and Hypothesis 2b was verified. Finally, the peer relationship of Tibetan adolescents was significantly positively associated with their perceived social support ($b = 0.51$, $p < 0.001$), and social support was also significantly positively associated with their prosocial behaviors ($b = 0.19$, $p < 0.001$). Moreover, the indirect effect of the peer relationship on the prosocial behavior of Tibetan adolescents via social support was significant ($b = 0.10$, $p < 0.001$, 95%CI = [0.046, 0.156]). In addition, after adding social support, peer relationships had no significant positive effect on prosocial behavior ($b = 0.04$, $p = 0.487$), indicating that Tibetan adolescents perceived social support mediated the relationship between their peer relationship and prosocial behavior, and Hypothesis 2c was verified. Together, Hypothesis 2 was verified.

2.3.3.6. The moderating effect of psychological resilience. Before testing the moderating effect, in order to reduce the possible risk of multicollinearity, this study standardized the independent variables and moderating variables respectively when constructing the interaction terms of independent variables and moderating variables. The test results are shown in Table 6. First, as shown in Model 4 of Table 6, the interaction term between parent–child relationship and psychological resilience had a significant positive effect on social support ($b = 0.13$, $p < 0.05$), indicating that psychological resilience moderated of parent–child relationship on social support, tentatively verifying Hypothesis 3a. Second, as shown in Model 5, the interaction term between teacher–student relationship and psychological resilience had a significant positive effect on social support ($b = 0.13$, $p < 0.05$), indicating that psychological resilience moderated of teacher–student relationship on social support, tentatively verifying the Hypothesis 3b. Finally, as shown in Model 6, the interaction term between peer relationship and psychological resilience had a significant positive effect on social support ($b = 0.11$, $p < 0.01$), indicating that psychological resilience moderated of peer relationship on social support, tentatively verifying Hypothesis 3c. To better visualize the moderating effect of psychological resilience, according to the suggestions of Cohen et al. (2013), the moderating effect was plotted at 1 SD above and 1 SD below the mean of psychological resilience, respectively. As shown in Figs. 2, 3, and 4, the simple slope results indicated that the parent–child, teacher–student, and peer relationships of Tibetan adolescents in the TAR had a stronger effect on social support when their psychological resilience levels were higher. Based on this, Hypothesis 3 was verified again.

2.3.3.7. Moderated mediating effect. In the moderated mediated effects test, this study used Bootstrapping method to estimate the confidence intervals for the indirect effects of parent–child, teacher–student, and peer relationships on the prosocial behavior of Tibetan adolescents via social support in contexts with high and low levels of psychological resilience, respectively, and the confidence intervals for the differences

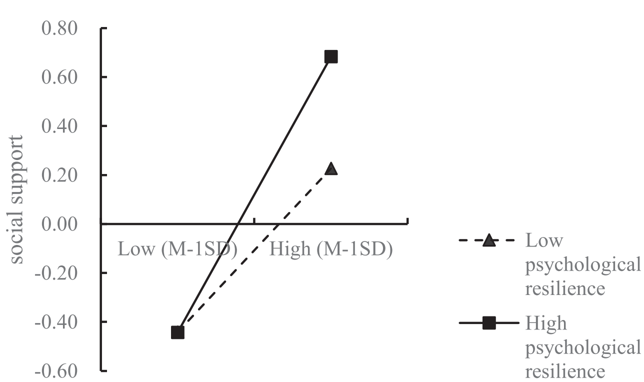

Fig. 2. The moderating effect of psychological resilience on the relationship between parent–child relationships and social support.

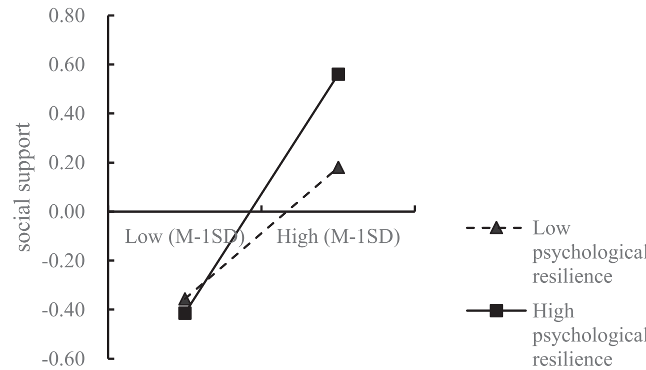

Fig. 3. The moderating effect of psychological resilience on the relationship between teacher–student relationships and social support.

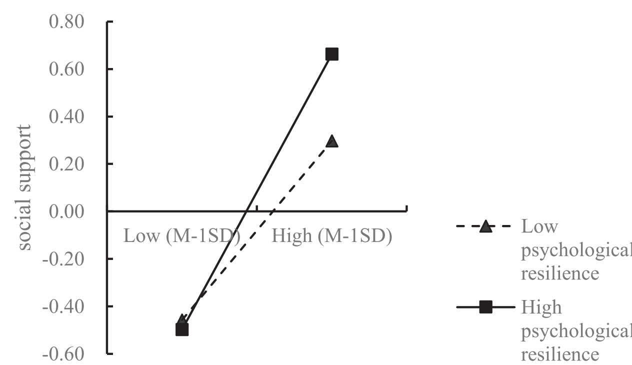

Fig. 4. The moderating effect of psychological resilience on the relationship between peer relationships and social support.

in indirect effects in contexts with high and low levels, and the results are shown in Table 8.

First, the indirect effect of parent–child relationships on prosocial behavior via social support was weaker at a low level of psychological

Table 8
Results of moderated mediating effect analysis.

Models	Prosocial behavior	Indirect effects	*SE*	95%CI
	Low level of psychological resilience	0.05	0.02	[0.023, 0.095]
Parent–child relationship → Social support → Prosocial behavior	High level of psychological resilience	0.10	0.03	[0.039, 0.150]
Indirect differences between high and low levels		0.04	0.02	[0.008, 0.081]
	Low level of psychological resilience	0.04	0.02	[0.016, 0.076]
Teacher–student relationship → Social support → Prosocial behavior	High level of psychological resilience	0.07	0.03	[0.028, 0.129]
Indirect differences between high and low levels		0.03	0.02	[0.007, 0.079]
	Low level of psychological resilience	0.06	0.02	[0.028, 0.110]
Peer relationship → Social support → Prosocial behavior	High level of psychological resilience	0.10	0.03	[0.042, 0.163]
Indirect differences between high and low levels		0.03	0.02	[0.007, 0.076]

Note: N = 544.

resilience ($b = 0.05$, 95%$CI = [0.023, 0.095]$) and stronger at a high level of psychological resilience ($b = 0.10$, 95%$CI = [0.039, 0.150]$), and the difference in indirect effect was significant between high and low levels of psychological resilience ($b = 0.04$, 95%$CI = [0.008, 0.081]$). Therefore, Hypothesis 4a was verified. Second, the indirect effect of teacher–student relationships on prosocial behavior via social support was weaker at a low level of psychological resilience ($b = 0.04$, 95%$CI = [0.016, 0.076]$) and stronger at a high level of psychological resilience ($b = 0.07$, 95%$CI = [0.028, 0.129]$), and the difference in indirect effect was significant between high and low levels of psychological resilience ($b = 0.03$, 95%$CI = [0.007, 0.079]$). Therefore, Hypothesis 4b was verified. Finally, the indirect effect of peer relationships on prosocial behavior via social support was weaker at a low level of psychological resilience ($b = 0.06$, 95%$CI = [0.028, 0.110]$) and stronger at a high level of psychological resilience ($b = 0.10$, 95%$CI = [0.042, 0.163]$), and the difference in indirect effect was significant between high and low levels of psychological resilience ($b = 0.03$, 95%$CI = [0.007, 0.076]$). Therefore, Hypothesis 4c was verified. Together, Hypothesis 4 was verified.

2.3.4. Study 2 summary

Study 2 verified that parent–child, teacher–student, and peer relationships among Tibetan adolescents had significant positive impacts on their prosocial behavior; this study also verified the mediating role played by social support in the connections between parent–child, teacher–student, and peer relationships, on the one hand, and prosocial behavior on the other. Simultaneously, the moderating effect analysis demonstrated that when psychological resilience was high, the relationship between parent–child, teacher–student, and peer relationships and social support was stronger, and psychological resilience positively moderated the mediating effect of social support.

3. Discussion

The main purpose of this study is to explore the influence of the main social relationships (parent–child, teacher–student, and peer relationships) of Tibetan adolescents on their prosocial behaviors as well as the mechanism underlying this influence. Through empirical analysis, this study confirmed that the main social relationships of Tibetan adolescents positively predict and promote their prosocial behavior. In addition, this study also compares the magnitude of the effects of the parent–child, teacher–student and peer relationships. The results indicate that teacher–student relationships had the greatest positive impact on the prosocial behavior of Tibetan adolescents, followed by peer relationships and parent–child relationships, the effects of which were relatively weak. The main reasons for this finding may be that, on the one hand, given the increasing autonomy associated with adolescence, Tibetan adolescents pay increasing attention to their primary relationships outside the family (De Goede, et al., 2009); on the other hand, the predominantly boarding school-based education and management system makes campus life much more noteworthy than family life for Tibetan adolescents, and teachers and peers are able to compensate for a lack of parental support (Moore et al., 2018), especially since high-quality teacher-student relationships grow to resemble a “safe base” for learning and growth, which helps Tibetan adolescents develop social skills and positive emotions, thereby stimulating their prosocial behavior (Endedijk et al., 2021; Obsuth et al., 2017). Furthermore, Quin (2017) also noted that high-quality teacher–student relationships can help overcome the disadvantages of family education, and the results of this study support this claim to a certain extent. In summary, the inclusion of the three main social relationships into the model comprehensively addresses the topic of the prediction and promotion of social relationships with regard to their effects on individual prosocial behavior, and the comparison of the magnitude of the effects of these three types of relationships provides a more targeted basis and suggestions for educational management and student growth.

Regarding the mediating role of social support, this study confirmed that the parent–child, teacher–student and peer relationships of Tibetan adolescents in the TAR have an effect on their prosocial behavior via their perceived social support. Social support theory claims that social support, as an external protective resource, is closely related to the quality and functioning of interpersonal relationships, and high-quality relationships can produce positive changes in an individual’s life and ultimately promote positive behavioral responses (Arslan, 2017; Hobfoll et al., 1990). For Tibetan adolescents, the social support provided by high-quality parent–child, teacher–student and peer relationships can not only provide resources that are necessary for their growth but also offer them a safe environmental climate that is conducive to activating and promoting their prosocial behavior (Izaguirre et al., 2021; Wang & Eccles, 2012).

This study confirmed the moderating effect of psychological resilience and found that when Tibetan adolescents’ psychological resilience was high, the positive effects of their parent–child, teacher–student and peer relationships on their social support were enhanced. Moreover, the indirect effects of their parent–child, teacher–student and peer relationships on their prosocial behavior via social support were stronger when psychological resilience was high. This finding suggests that Tibetan adolescents with higher levels of psychological resilience may have more internal resources (e.g., optimism, self-reliance, and self-efficacy) and external resources (e.g., social support), thus enabling them to pay more attention to the plight and needs of others and promoting their prosocial behavior.

3.1. Implications

This study provides a rich body of research and empirical findings to support educational development and research in western China. During the investigation conducted for this study, the research team also found that many Tibetan adolescents would choose to remain in the TAR to continue living and working after finishing their middle school studies. Therefore, it is of great theoretical and practical significance for this study to select Tibetan adolescents as its research object.

In terms of theoretical significance, first, this study enriches the literature concerning “social relationships-positive behavior” from the perspective of social support theory. Previous studies on this topic have mostly been based on attachment theory, social exchange theory, social learning theory, or self-determination theory (Froiland et al., 2019; Roorda et al., 2011; van Hoorn et al., 2016; Yoo et al., 2013). This study regards the parent–child, teacher–student, and peer relationships as individual resources and explores the influence of these three main types of social relationships on the prosocial behavior of Tibetan adolescents based on social support theory. Second, by incorporating multiple social relationships into the model simultaneously, this study provides a more comprehensive answer to the question of how social relationships are able to predict individual prosocial behavior. When discussing the relationship between social relationships and prosocial behavior, previous studies have mostly focused on a certain type (class) of relationship and have not examined the impact of multiple social relationships on the prosocial behavior of adolescents comprehensively from the perspective of relationship integration. Third, this study reveals the boundary conditions associated with Tibetan adolescents’ parent–child, teacher–student, and peer relationships as mediated by social support, which affects their prosocial behavior. It is noteworthy that the current academic community has little understanding of the boundary conditions under which the primary social relationships of adolescents affect their prosocial behaviors. Indeed, the development of prosocial behavior in adolescents is more closely associated with the interaction between individual personality traits and the environment (Carlo & Padilla-Walker, 2020). Simultaneously, this study responds to the suggestion by previous scholars that future research should explore the ways in which personality traits affect perceptions or acceptance of social support and improve our understanding of psychological resilience (Thoits, 1995).

In terms of practical significance, first, the main social relationships of Tibetan adolescents in the TAR are unique. These significant differences are not only the result of the unique cultural background and geographical environment of the TAR but also of the growth experiences and educational management of Tibetan adolescents. With regard to their actual family education and school education, the focus on Tibetan adolescents' self, group and development should be increased continuously, and students should be guided to establish positive parent–child, teacher–student and peer relationships to continuously activate the linkage from social relationships to students' positive behaviors, thereby stimulating and promoting the education of students who exhibit more positive attitudes and behaviors.

Second, according to the results of the study, the primary social relationships of Tibetan adolescents all have positive predictive and promotional effects on the prosocial behaviors of those students, especially the teacher–student relationship, which plays the greatest role in this process, followed by the peer relationship, with the parent–child relationship being the weakest. From this perspective, the role of teacher–student relationships in guiding students' positive behaviors should be emphasized. In addition to encouraging teachers to strengthen the construction of their own morality and style and to begin with their own "upright mind and self-improvement", teachers should be encouraged to grow closer to students and focus on "cultivating people through virtue" by offering ideological and political courses and providing ideological and political guidance. Moreover, for peer relationships, at the macro (school management) level, an atmosphere of peer motivation should be created, and at the micro (class management) level, more peer role models should be encouraged, more peer examples should be promoted, and more healthy relationships should be established. Additionally, regarding the parent–child relationship, the research team found that the parents of most Tibetan adolescents in the TAR are farmers and herdsmen with low levels of education, and there are even some families in which neither parent has ever left the village, which greatly limits the parents' influence on their children's exposure and contributions to society. Therefore, based on the process by which the parent–child relationship affects students' prosocial behavior, students can be encouraged to pass more fresh, cutting-edge and advanced knowledge and information on to their parents, and teachers can be encouraged to actively conduct home visits to communicate the care of the school and society, thus promoting ideological progress and improving the level and quality of parent–child care with respect to the local students' parents through joint efforts.

Finally, students' prosocial behavior can be activated and promoted by enhancing their perceptions of social support and psychological resilience. Care and support from their families, respect and trust from their peers, and help and encouragement from teachers as well as guidance and advice from schools, assistance and resources from society, recognition and funding from other organizations, all of which can improve students' perceptions of emotional, informational, instrumental, and belonging support to some degree with the aim of improving the prosocial behavior of Tibetan adolescents (Cohen & Wills, 1985). In addition, the improvement of psychological resilience can also influence students' positive behaviors. Therefore, in the process of education and teaching, it is necessary to encourage and assist students in the task of developing a high level of psychological resilience, such as an unyielding spirit and an attitude of continuous struggle.

3.2. *Limitations and Future Directions*

This study provided evidence of the relationship between the main social relationships of Tibetan adolescents and their prosocial behavior in the TAR. However, there remain certain directions for future research that could strive to improve our understanding of this topic. First, this study is a cross-sectional study and is thus limited to reflecting statistically significant effects; accordingly, this study faces certain limitations with regard to making inferences regarding the causal relationships between the main social relationships and prosocial behavior of Tibetan adolescents. Future studies can explore the specific relationships among these variables in further detail by using longitudinal studies or experimental designs. Second, this study controlled only for the gender and age of Tibetan adolescents, and future studies could consider other relevant control variables, such as the students' grades or their parents' levels of education. Third, this study used psychological resilience as a moderating variable to investigate differences in individual personality traits; however, psychological resilience is a multidimensional structure that includes characteristics such as resilience, optimism, self-efficacy, and goal orientation. Future research could investigate the aspects of psychological resilience that are most significant with respect to moderating the relationship between the primary social relationships and social support of Tibetan adolescents in further detail and thus provide additional insight into ways of promoting the prosocial behavior of Tibetan adolescents.

4. Conclusions

This study takes Tibetan adolescents in the TAR as its research object and uses the theory of social support to explore the influence of the main social relationships (parent–child, teacher–student, and peer relationships) of Tibetan adolescents on their prosocial behaviors as well as the mechanism underlying this influence. The results show that: First, the main social relationships of Tibetan adolescents differ from those of local Han adolescents and Han adolescents in mainland China; Second, the parent–child, teacher–student and peer relationships of Tibetan adolescents are significantly positively correlated with their prosocial behavior, and according to the data results, the teacher–student relationship has the greatest impact on the prosocial behavior of Tibetan adolescents, followed by peer relationships, with the parent–child relationship being the weakest; Third, the parent–child, teacher–student and peer relationships of Tibetan adolescents have an effect on their prosocial behavior via their perceived social support; Fourth, the psychological resilience of Tibetan adolescents positively moderates the relationship between their parent–child, teacher–student, and peer relationships and social support, while psychological resilience has a positive moderating effect on the mediating effect of social support.

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CRediT authorship contribution statement

Ming Kong: Conceptualization, Data curation, Funding acquisition, Investigation, Supervision. **Yahua Lu:** Methodology, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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